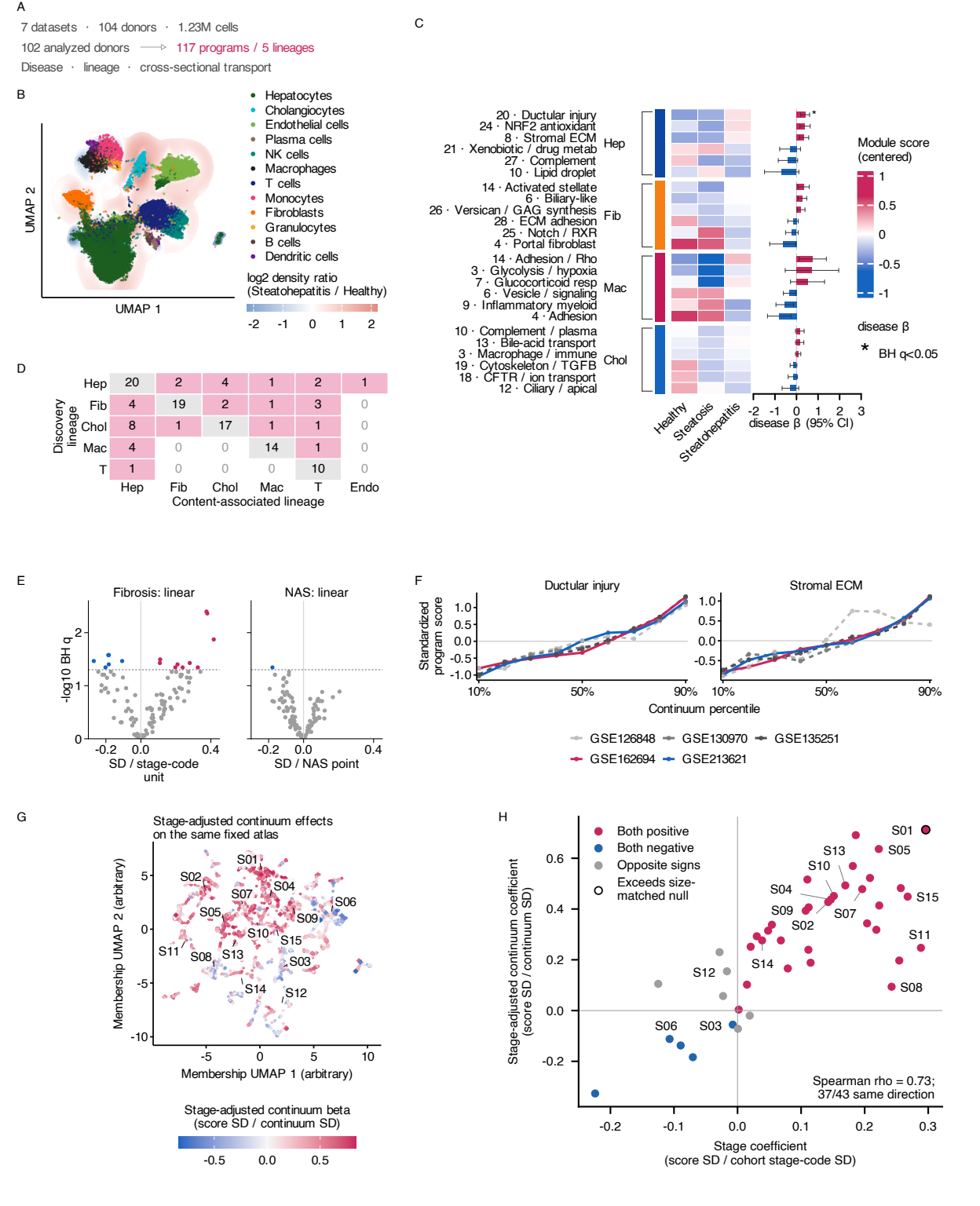

A
7 datasets · 104 donors · 1.23M cells
102 analyzed donors
117 programs / 5 lineages
Disease · lineage · cross-sectional transport
B
UMAP 2
UMAP 1
Hepatocytes
Cholangiocytes
Endothelial cells
Plasma cells
NK cells
Macrophages
T cells
Monocytes
Fibroblasts
Granulocytes
B cells
Dendritic cells
log2 density ratio
(Steatohepatitis / Healthy)
-2 -1 0 1 2
C
20 · Ductular injury
24 · NRF2 antioxidant
8 · Stromal ECM
21 · Xenobiotic / drug metab
27 · Complement
10 · Lipid droplet
14 · Activated stellate
6 · Biliary-like
26 · Versican / GAG synthesis
28 · ECM adhesion
25 · Notch / RXR
4 · Portal fibroblast
14 · Adhesion / Rho
3 · Glycolysis / hypoxia
7 · Glucocorticoid resp
6 · Vesicle / signaling
9 · Inflammatory myeloid
4 · Adhesion
10 · Complement / plasma
13 · Bile-acid transport
3 · Macrophage / immune
19 · Cytoskeleton / TGFB
18 · CFTR / ion transport
12 · Ciliary / apical
Hep
Fib
Mac
Chol
Healthy
Steatosis
Steatohepatitis
-2 -1 0 1 2 3
disease β (95% CI)
Module score
(centered)
1
0.5
0
-0.5
-1
disease β
* BH q<0.05
D
Discovery lineage
Hep 20 2 4 1 2 1
Fib 4 19 2 1 3 0
Chol 8 1 17 1 1 0
Mac 4 0 0 14 1 0
T 1 0 0 0 10 0
Hep Fib Chol Mac T Endo
Content-associated lineage
E
Fibrosis: linear
NAS: linear
-log10 BH q
0 1 2
-0.2 0.0 0.4
SD / stage-code unit
SD / NAS point
F
Ductular injury
Stromal ECM
Standardized program score
1.0 0.5 0.0 -0.5 -1.0
10% 50% 90%
Continuum percentile
GSE126848
GSE130970
GSE135251
GSE162694
GSE213621
G
Stage-adjusted continuum effects on the same fixed atlas
Membership UMAP 2 (arbitrary)
Membership UMAP 1 (arbitrary)
5 0 -5 -10
-5 0 5 10
S01
S02
S04
S07
S05
S06
S09
S10
S15
S11
S08
S13
S03
S14
S12
Stage-adjusted continuum beta
(score SD / continuum SD)
-0.5 0.0 0.5
H
Both positive
Both negative
Opposite signs
Exceeds size-matched null
Stage-adjusted continuum coefficient (score SD / continuum SD)
0.6 0.4 0.2 0.0 -0.2
-0.2 -0.1 0.0 0.1 0.2 0.3
Stage coefficient
(score SD / cohort stage-code SD)
Spearman rho = 0.73;
37/43 same direction
S01
S13
S05
S10
S04
S15
S09
S07
S02
S11
S14
S12
S08
S06
S03